YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 1

Financial applications

Question 1

Ava gets $2000 for her 18th birthday and wants to deposit it into a high-interest account while she thinks about how to spend it. Her current bank offers two options:

- Simple Savings account: 9.75% p.a.
- Clever Compounds account: 7.99% p.a. (compounding yearly).

a How much interest would Ava earn in a year with each account?

2 marks

b If Ava left the money in the account for 3 years, what would be the balance of each bank account?

2 marks

c Complete the table below to determine how many years it would take for the Clever Compounds account to become more beneficial than the Simple Savings account.

Year	Simple Savings account ($)	Clever Compounds account ($)
0	2000	2000

3 marks

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Another bank offers a compound interest account, Monthly Savings, earning 7.5% compounding monthly.

d How much interest would Ava earn in the first year if she chose this account?

2 marks

Ava's friend, Isaac, tells her about the savings account, Savings Quarterly, that he has been using for the past 3 years. He can't remember the interest rate, but he knows that he started with a deposit of $10 000, and after 3 years of compounding quarterly he now has $12 534.01.

e Find the interest rate of Isaac's savings account.

2 marks

f If she chose this account, what would be Ava's account balance at the end of 3 years?

1 mark

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g Draw a single graph showing the growing balance for these four savings accounts over 3 years.

3 marks

h If Ava wanted to deposit her money for 10 years, which account would you recommend to her?

3 marks

Total 18 marks

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Question 2

Finn goes shopping for a new laptop. He sees one he wants for $1699 at Lappys, but cannot afford that much up front. The store offers him a deal where he pays a 10% deposit, and then $79 a month for 2 years.

a How much interest will Finn pay if he accepts this deal?

2 marks

b What is the flat rate of interest charged per annum on the laptop?

2 marks

The competing store, Technica, offers the same laptop for $1850 under a 'Buy now, pay later' deal. Finn was able to take home the laptop that day and pay nothing for 14 months. Any outstanding balance will be charged at an interest rate of 5% per month from the date of purchase.

c How much will Finn need to pay per month towards this laptop if he is to pay it off before the 14 months are up?

1 mark

d If Finn can pay the full $1850 before the 14 months are up, should he take this deal or the one from Lappys?

1 mark

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e Finn is able to put $110 per month towards the laptop. How much will he still owe just before the end of the 14 months?

1 mark

f Calculate the interest that will be charged on the remaining balance and, hence, state what Finn will now owe when the 14 months are up.

1 mark

g Copy and complete the table below to show how long it will take Finn to pay back the loan if he continues to put $110 a month towards the laptop. Assume that Finn pays $100 at the start of the month and that the interest is calculated at the end of the month.

Month	Balance ($)	Minus payment ($)	Balance after payment ($)	Plus interest ($)	End balance ($)
14		110			
15		110			

4 marks

h Use the table from part **g** to show which deal Finn should take.

2 marks

Total 14 marks

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Question 3

Olivia purchases a new house for $575 000. She has $80 000 for the deposit and must borrow the remaining balance from the bank. Her bank offers her a 30-year loan with an interest rate of 6.75% compounding monthly.

a What will be Olivia's monthly repayments?

1 mark

b How much interest will Olivia pay over the life of the loan?

2 marks

After 10 years, Olivia is able to raise her monthly repayments to $4000.

c How many months will Olivia save on her loan if she pays $4000 a month for the remainder of the loan?

3 marks

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d How much interest will she save?

2 marks

After a further 5 years, the bank raises the interest rate to 7.39%.

e Calculate how much interest Olivia will now pay over the life of the loan.

4 marks

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After a total of 20 years, Olivia needs to undertake $100 000 worth of renovations on the house, and she redraws on her home loan in order to do so.

f Find her monthly repayments if she chooses to keep the original 30-year term.

4 marks

g If she keeps paying $4000 a month, calculate the:

i total amount of time taken to pay off the loan

2 marks

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ii total amount of interest she pays on the loan.

2 marks

h If Olivia increases her monthly repayments to $10 000, can she limit the interest she pays to $600 000?

3 marks
Total 23 marks

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Question 4

Five years ago, Alex opened an account with an interest rate of 4.95%, compounding monthly. At the beginning of every month he deposited $500.

a Calculate his current account balance.

1 mark

Alex manages to double his deposit amount for the next 5 years.

b Calculate the account balance at the end of 10 years.

1 mark

c Find how much more interest Alex has earned by doubling his deposit amount.

3 marks

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At the end of 10 years, Alex is injured on the job, and is unable to work for 2 years. He is compensated but, after medical bills, he is unable to continue depositing money into his savings account. He returns to work on a part-time basis for the following 3 years, and is able to deposit $500 a month for those 3 years.

d Find the value of Alex's account 15 years after it was opened.

2 marks

At the end of the 15 years, Alex decides to use the savings as a perpetuity. He switches the money to an investment that pays 5.25% p.a., compounding quarterly.

e How much will he receive each quarter from this perpetuity?

1 mark

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Alex finds that this isn't quite as much money as he would like.

f If he decides to withdraw $6000 per quarter, how many years will the money last?

1 mark

g If Alex wants the money to last 10 years, to what amount must he limit his withdrawals every quarter?

1 mark

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h How much longer would the annuity last if, after 5 years, the interest rate increases to 6.9% p.a.? Assume that Alex is withdrawing the amount you calculated in part **g**.

3 marks

i Find how much interest Alex will have earned from the account from the time that he opened the account until he drained it.

2 marks

Total 15 marks

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